1 Solving for Nash Equilibrium

In this section, we develop the methodology for computing a Nash equilibrium of LCP. While LCP has many Nash equilibria, we choose to study a specific one with several nice properties. Our approach proceeds in three stages: (1) establishing desired properties of the equilibrium, (2) characterizing the structure that such a Nash equilibrium must satisfy, and (3) deriving a system of equations whose solution yields the equilibrium strategy profile. The complete derivation and verification that our solution constitutes a Nash equilibrium is provided in Appendix **??**.

1.1 Equilibrium Selection

Like FBCP, LCP has an infinite class of Nash equilibria. For the purposes of this paper, we construct a specific Nash equilibrium which align with intuition and interpretability. Specifically, we find a Nash equilibrium where strategies can be represented as continuous functions where possible (e.g. mapping hand strengths to bet sizes). Second, we look for a *monotone* calling strategy (defined below). Finally, we choose a betting strategy which orders bluffs in weakly decreasing size, motivated by a less rigorous but intuitive argument. These refinements are not strictly necessary, but give us an equilibrium which aligns well with traditional poker intuition and is easy to describe mathematically.

1.2 Monotone Strategies

Definition 1.1 (Monotone Calling Strategy)**.** A *monotone* calling strategy is a pure strategy such that for any bet size s and any two hand strengths $y_1 < y_2$, if the caller calls a bet of size s with y_1, they must also call with y_2.

This should sound intuitive. Violating this condition (in a non-negligible way) is actually weakly dominated. Not only is a monotone strategy weakly better against all opponents, but there exists an opponent against which the monotone strategy is strictly better (see Appendix **??** for proof). Restricting to pure strategies can be explained similarly: it is better to always call with a stronger hand and always fold a weaker one than to mix between the two.

This gives us a way to narrow the space of possible calling strategies. But what about betting strategies? We mentioned earlier that many optimal betting strategies differ only in how they bluff. The bettor always bluffs with their weakest hands. They will have some threshold hand strength such that every hand below this will bluff, but the sizes of these bluffs and ordering of the sizes within this hand strength region are unprescribed. When the caller plays optimally, the caller never calls with a hand below this bluffing threshold, so they never lose to a bluff. However, if the caller deviates to a suboptimal strategy, then the bettor's bluffing hand strength might come into play. We need to make a certain bluff size a certain proportion of the time to balance our value bets, but how should we order these bluffs?

Conventional poker wisdom dictates that the bettor should make their largest bluffs with their weakest hands, and their smaller bluffs with the stronger of the (still weak) bluffing hands[**?**]. It turns out that there is a mathematical justification for this intuition, but that it makes many assumptions about the caller's strategy. In traditional poker, we generally expect that the caller will either call with the optimal frequency, 'overcall' (call with weaker hands than they should), or 'undercall' (fold hands they should call with). If the caller overcalls, and if they are more likely to call smaller bets (another strong assumption), then the bettor might 'accidentally' win a showdown when bluffing. However, this is far more likely to happen if the bettor makes their small bluffs with their strongest bluffing hands. If they make their small bluffs with their strongest hands, then they never win these overcalling situations. We will not try to formalize this argument because it relies on too many restrictions on the caller's strategy, but it does provide justification for why this sort of bluffing strategy is desirable.

Thus, we choose to analyze a Nash equilibrium in which the bettor orders their bluffs in non-increasing size as a function of hand strength (largest bluffs with weakest hands). Keep in mind that we will formally prove that the constructed strategy profile is a Nash equilibrium–this discussion is purely motivation for the specific choice of equilibrium.

1.3 Nash Equilibrium Structure

We will now describe the structure of the Nash equilibrium in terms of thresholds x_i and functions $c(s)$, $b(s)$, and $v(s)$. These turn out to be fully determined by the parameters L and U, but for now they are unknown. Notice that both players use pure strategies, like in NLCP: the bettor maps hand strengths directly to bet sizes, and the caller maps hand strengths and bet sizes to actions with no mixing. We can break strategies into piecewise regions as follows (see figure **??** for a visual representation):

- The caller has a calling threshold $c(s)$ that is continuous and differentiable in s, including at endpoints L and U. They call with hands $y \geq c(s)$ and fold with hands $y < c(s)$[1].

[1]The action taken at the threshold is irrelevant, since it occurs with probability zero.

- The bettor partitions $[0, 1]$ into three regions: bluffing $x \in [0, x_2]$, checking $x \in [x_2, x_3]$, and value betting $x \in [x_3, 1]$.
- Within the bluffing region, the bettor partitions into a max-betting region $x \in [0, x_0]$, an intermediate region $x \in [x_0, x_1]$, and a min-betting region $x \in [x_1, x_2]$ (this is using the assumption of non-increasing bluffing sizes mentioned above).
- Within the intermediate bluffing region, the bettor bets according to a continuous, decreasing function $s = b^{-1}(x)$ with endpoints $b^{-1}(x_0) = U$ and $b^{-1}(x_1) = L$. Note that $b(s)$ gives hand strength as a function of bet size, which feels backwards, but turns out to be more mathematically convenient and still equally descriptive.
- Within the value betting region, the bettor partitions into a min-betting region $x \in [x_3, x_4]$, an intermediate region $x \in [x_4, x_5]$, and a max-betting region $x \in [x_5, 1]$.
- Within the intermediate value betting region, the bettor bets according to a continuous, increasing function $s = v^{-1}(x)$ with endpoints $v^{-1}(x_4) = L$ and $v^{-1}(x_5) = U$. Once again, $v(s)$ gives hand strength as a function of bet size, which feels backwards but is more mathematically convenient.

1.4 Constraints and Indifference Equations

Having established the qualitative structure of the Nash equilibrium, we now derive the quantitative relationships that the strategy profile must satisfy. These constraints arise from two fundamental equilibrium conditions: players must be indifferent among actions at exact thresholds, and players must optimize when choosing among available actions. The resulting system of differential and algebraic equations will uniquely determine all threshold values $x_0, \ldots, x_5$ and the functions $b(s)$, $v(s)$, and $c(s)$, which in turn uniquely determine the strategy profile.

The key conditions are:

- The caller must be indifferent between calling and folding at their calling threshold
- The bettor must be indifferent between checking and betting at their value betting and bluffing thresholds
- The bettor's bet size for a value bet must maximize their expected value

- The bettor's strategy must be continuous in bet size (in the regions where they bet)

These conditions give us the following system of equations, which we will derive in the next section:

Caller Indifference:

$$(x_2 - x_1) \cdot (1 + L) - (x_4 - x_3) \cdot L = 0 \tag{1}$$
$$x_0 \cdot (1 + U) - (1 - x_5) \cdot U = 0 \tag{2}$$
$$|b'(s)| \cdot (1 + s) - |v'(s)| \cdot s = 0 \tag{3}$$

Bettor Indifference and Optimality:

$$-sc'(s) - c(s) + 2v(s) - 1 = 0 \tag{4}$$
$$(x_3 - c(L)) \cdot (1 + L) - (1 - x_3) \cdot (L) + c(L) = x_3 \tag{5}$$
$$c(s) - (1 - c(s)) \cdot s = x_2 \tag{6}$$

Continuity Constraints:

$$b(U) = x_0 \tag{7}$$
$$b(L) = x_1 \tag{8}$$
$$v(U) = x_5 \tag{9}$$
$$v(L) = x_4. \tag{10}$$

Note that for this analysis, it is simpler to pretend that payoffs exclude the initial ante of 0.5, since this is a sunk cost to both players and optimization only depends on relative payoffs between actions.

1.4.1 Caller Indifference

By definition, $c(s)$ is the threshold above which the caller calls and below which they fold. This means that in Nash Equilibrium, the caller must be indifferent between calling and folding with a hand strength of $c(s)$:

$$\mathbb{E}[\text{call } c(s)] = \mathbb{E}[\text{fold } c(s)]$$
$$\mathbb{P}[\text{bluff}|s] \cdot (1 + s) - \mathbb{P}[\text{value bet}|s] \cdot s = 0.$$

We now split into cases based on the value of s.

Case 1: $s = L$. The hands the bettor value bets L with are $x \in (x_3, x_4)$, and the hands they bluff with are $x \in (x_1, x_2)$.

$$(x_2 - x_1) \cdot (1 + L) - (x_4 - x_3) \cdot L = 0. \tag{11}$$

Here, we are implicitly multiplying both sides by the common denominator of $(x_4 - x_3) + (x_2 - x_1)$.

Case 2: $s = U$. The hands the bettor value bets U with are $x \in (x_5, 1)$, and the hands they bluff with are $x \in (0, x_0)$.

$$(1 - x_5) \cdot (1 + U) - x_0 \cdot U = 0, \tag{12}$$

again, implicitly multiplying both sides by the common denominator of $(1 - x_5) + x_0$.

Case 3: $L \leq s \leq U$. In this case, the bettor has exactly one value hand and one bluffing hand, but somewhat paradoxically, they are not equally likely. The probability of a value bet given the size s is related to the inverse derivative of the value function $v(s)$ at s, and the same goes for a bluff. This gives us the following relation:

$$\frac{\mathbb{P}[\text{value bet}|s]}{\mathbb{P}[\text{bluff}|s]} = \frac{|b'(s)|}{|v'(s)|}$$

An intuitive interpretation of this is that for any small neighborhood around the bet size s, the bettor has more hands which use a bet size in the neighborhood if $v(s)$ does not change rapidly around s, that is, if $|v'(s)|$ is small. The same goes for bluffing hands, and as we limit the neighborhood to a single point, the ratio of the two probabilities approaches the ratio of the derivatives. We know that these are the only two possible bettor actions for such a bet size, so

$$\mathbb{P}[\text{value bet}|s] = \frac{|b'(s)|}{|b'(s)| + |v'(s)|}$$

$$\mathbb{P}[\text{bluff}|s] = \frac{|v'(s)|}{|b'(s)| + |v'(s)|}$$

Plugging this into the indifference equation and dividing out the common denominator, we get:

$$|b'(s)| \cdot (1 + s) - |v'(s)| \cdot s = 0. \tag{13}$$

1.4.2 Bettor Indifference and Optimality

When the bettor makes a value bet, they are attempting to maximize the expected value of the bet. We can write the expected value of a value bet as:

$$\begin{aligned}\mathbb{E}[\text{value bet } s|x] &= \mathbb{P}[\text{call with worse}] \cdot (1 + s) - \mathbb{P}[\text{call with better}] \cdot s + \mathbb{P}[\text{fold}] \cdot 1 \\ &= (x - c(s)) \cdot (1 + s) - (1 - x) \cdot (s) + c(s).\end{aligned}$$

To maximize this, we take the derivative with respect to s and set it equal to zero. Crucially, we are treating $c(s)$ as a function of s and using the chain rule, since changing the bet size s will also change the calling threshold $c(s)$. We want this optimality condition to hold for the bettor's Nash equilibrium strategy, so we set $x = v(s)$. This gives us:

$$
\begin{aligned}
&\frac{d}{ds}\mathbb{E}[\text{value bet } s|x = v(s)] = 0 \\
&-sc'(s) - c(s) + 2v(s) - 1 = 0.
\end{aligned} \tag{14}
$$

Additionally, when the bettor has the most marginal value betting hand at $x = x_3$, they should be indifferent between a minimum value bet and a check:

$$
\begin{aligned}
\mathbb{E}[\text{value bet } L|x = x_3] &= \mathbb{E}[\text{check}|x = x_3] \\
(x_3 - c(L)) \cdot (1 + L) - (1 - x_3) \cdot (L) + c(L) &= x_3.
\end{aligned} \tag{15}
$$

Finally, when the bettor has the most marginal bluffing hand at $x = x_2$, they should be indifferent between a minimum bluff and a check. However, as we discussed earlier, the bettor should be indifferent among all bluffing sizes, so the bettor should actually be indifferent between checking and making any bluffing size s at $x = x_2$. This gives us:

$$
\begin{aligned}
\mathbb{E}[\text{bluff } s|x = x_2] &= \mathbb{E}[\text{check}|x = x_2] \\
c(s) - (1 - c(s)) \cdot s &= x_2.
\end{aligned} \tag{16}
$$

1.4.3 Continuity Constraints

As discussed above, the bettor's strategy is continuous in s and x (except when checking). This means that the endpoints of the functions $v(s)$ and $b(s)$ are constrained as follows:

$$
b(U) = x_0, \quad b(L) = x_1, \quad v(U) = x_5, \quad v(L) = x_4. \tag{17}
$$